

3.8 Use Case Specifications

3.8.1 Use Case UC-001 – User Login

Field	Description
Use Case ID	UC-001
Use Case Name	User Login
Primary Actor	Admin / Customer Service Agent / Supervisor / Retention Specialist
Description	Users log into the Telecom CRM System using valid credentials
Preconditions	User account must exist and be active.
Trigger	User enters login credentials.
Main Flow	1. User enters email and password. 2. System validates credentials. 3. System authenticates user. 4. User redirected to dashboard.
Alternative Flow	Invalid credentials → Error message displayed
Postconditions	User gains authorized system access
Business Rules	Role-Based Access Control (RBAC) applies.
Acceptance Criteria	User logs in successfully and accesses authorized modules.

3.8.2 Use Case UC-002 – Search Customer

Field	Description
Use Case ID	UC-002
Use Case Name	Search Customer
Primary Actor	Customer Service Agent
Description	Agent searches for a telecom customer using available identifiers.
Preconditions	Agent must be logged in.
Trigger	Agent initiates customer search.
Main Flow	1. Agent enters National ID / Phone Number / Customer Name. 2. System searches customer database. 3. Matching customer profile displayed.
Alternative Flow	No customer found → System displays error message.
Postconditions	Customer profile becomes accessible.
Business Rules	Search should support partial matching
Acceptance Criteria	Correct customer information displayed

3.8.3 Use Case UC-003 – Create Complaint Ticket

Field	Description
Use Case ID	UC-003
Use Case Name	Create Complaint Ticket
Primary Actor	Customer Service Agent
Description	Agent creates a complaint ticket for a telecom customer.
Preconditions	Customer profile must exist.
Trigger	Customer submits complaint.
Main Flow	1. Agent searches customer. 2. Agent selects complaint category. 3. Agent enters complaint details. 4. System generates Ticket ID. 5. Complaint saved successfully.
Alternative Flow	No customer found → System displays error message.
Postconditions	Complaint ticket created successfully.
Business Rules	Ticket Priority = Low / Medium / High.
Acceptance Criteria	Complaint created and status defaults to “Open”

3.8.4 Use Case UC-004 – Complaint Escalation

Field	Description
Use Case ID	UC-004
Use Case Name	Complaint Escalation
Primary Actor	Supervisor / n8n Automation
Description	Complaint escalates automatically if SLA time exceeded.
Preconditions	Complaint must remain unresolved beyond SLA
Trigger	SLA timer exceeded.
Main Flow	1. n8n detects SLA breach. 2. System escalates complaint. 3. Supervisor receives notification.
Alternative Flow	Complaint resolved before SLA → No escalation.
Postconditions	Complaint status becomes “Escalated”.
Business Rules	High-priority complaints escalate immediately.
Acceptance Criteria	Supervisor notified successfully

3.8.5 Use Case UC-005 – Renew Package

Field	Description
Use Case ID	UC-005
Use Case Name	Renew Package
Primary Actor	Customer Service Agent
Description	Agent renews customer telecom package
Preconditions	Customer account active
Trigger	Customer requests renewal.
Main Flow	1. Agent opens customer profile. 2. Agent selects renewal option. 3. System validates eligibility. 4. Package renewed.
Alternative Flow	Billing issue → Renewal rejected
Postconditions	Package status updated.
Business Rules	Only active accounts can renew.
Acceptance Criteria	Renewal confirmation displayed.

3.8.6 Use Case UC-006 – View Customer Profile

Field	Description
Use Case ID	UC-006
Use Case Name	View Customer Profile
Primary Actor	Customer Service Agent / Supervisor
Description	Users view customer information, active subscriptions, billing, and complaint history.
Preconditions	Customer must exist in the system.
Trigger	User selects customer profile.
Main Flow	1. User searches customer. 2. System retrieves profile data. 3. Customer information displayed.
Alternative Flow	Customer not found → Error message displayed.
Postconditions	Customer details become accessible.
Business Rules	Only authorized users can access customer profile.
Acceptance Criteria	Profile data displays accurately.

3.8.7 Use Case UC-007 – Assign Complaint

Field	Description
Use Case ID	UC-007
Use Case Name	Assign Complaint
Primary Actor	Supervisor
Description	Supervisor assigns complaint ticket to available agent.
Preconditions	Complaint must exist and status = Open.
Trigger	Supervisor opens complaint queue.
Main Flow	1. Supervisor selects complaint. 2. Supervisor selects available agent. 3. System updates assignment. 4. Agent receives notification.
Alternative Flow	No available agent → Queue remains pending.
Postconditions	Complaint assigned successfully.
Business Rules	High-priority complaints should be assigned first.
Acceptance Criteria	Assigned agent receives notification.

3.8.8 Use Case UC-008 – Update Complaint Status

Field	Description
Use Case ID	UC-008
Use Case Name	Update Complaint Status
Primary Actor	Customer Service Agent / Supervisor
Description	Users update complaint progress.
Preconditions	Complaint must exist.
Trigger	Agent updates ticket progress.
Main Flow	1. User opens complaint. 2. User selects status. 3. System saves update.
Alternative Flow	Invalid update → Validation message.
Postconditions	Complaint status updated.
Business Rules	Allowed statuses: Open, In Progress, Escalated, Resolved, Closed.
Acceptance Criteria	Complaint timeline updated correctly.

3.8.9 Use Case UC-009 – Upgrade/Downgrade Package

Field	Description
Use Case ID	UC-009
Use Case Name	Upgrade/Downgrade Package
Primary Actor	Customer Service Agent
Description	Agent changes customer subscription package.
Preconditions	Customer account active.
Trigger	Customer requests package change.
Main Flow	1. Agent opens package details. 2. Agent selects new package. 3. System validates eligibility. 4. Package updated.
Alternative Flow	Package unavailable → Error displayed.
Postconditions	Subscription updated successfully.
Business Rules	Eligibility rules apply.
Acceptance Criteria	Package changes reflect immediately.

3.8.10 Use Case UC-010 – View Billing History

Field	Description
Use Case ID	UC-010
Use Case Name	View Billing History
Primary Actor	Customer Service Agent / Supervisor
Description	Users view customer billing history and due balance.
Preconditions	Customer profile exists.
Trigger	User selects billing section.
Main Flow	1. User opens billing tab. 2. System retrieves billing history. 3. Payment records displayed.
Alternative Flow	No billing data → Empty state shown.
Postconditions	Billing history visible.
Business Rules	Billing data must be real-time.
Acceptance Criteria	Accurate billing details displayed.

3.8.11 Use Case UC-011 – Generate Complaint Report

Field	Description
Use Case ID	UC-011
Use Case Name	Generate Complaint Report
Primary Actor	Supervisor
Description	Supervisor generates complaint performance report.
Preconditions	Complaint data exists.
Trigger	SLA timer exceeded.
Main Flow	1. Supervisor selects report filters. 2. System fetches complaint KPIs. 3. Report generated. 4. Export/download available.
Alternative Flow	No data available → Empty report displayed.
Postconditions	Report generated successfully.
Business Rules	Reports support export formats (PDF/Excel).
Acceptance Criteria	Report generated without errors.

3.8.12 Use Case UC-012 – Package Expiry Reminder (n8n)

Field	Description
Use Case ID	UC-012
Use Case Name	Package Expiry Reminder
Primary Actor	n8n Automation
Description	n8n automatically sends package expiry reminders to customers.
Preconditions	Package expires within 3 days.
Trigger	Daily scheduled cron trigger.
Main Flow	1. n8n checks expiring packages. 2. System validates active subscriptions. 3. SMS/Email reminder sent. 4. CRM updated.
Alternative Flow	Customer inactive → Reminder skipped.
Postconditions	Reminder sent successfully.
Business Rules	No duplicate reminders.
Acceptance Criteria	Customer notified before expiry.

3.8.13 Use Case UC-013 – Weekly KPI Report Automation (n8n)

Field	Description
Use Case ID	UC-013
Use Case Name	Weekly KPI Report Automation
Primary Actor	n8n Automation
Description	n8n automatically generates KPI reports and emails management.
Preconditions	Weekly schedule active.
Trigger	Weekly scheduler execution.
Main Flow	1. n8n fetches CRM KPIs. 2. System generates report. 3. Report emailed to management.
Alternative Flow	KPI unavailable → Log error.
Postconditions	Weekly report delivered.
Business Rules	Reports generated weekly.
Acceptance Criteria	Management receives accurate KPI report.

3.8.14 Use Case UC-014 – Weekly KPI Report Automation (n8n)

Field	Description
Use Case ID	UC-014
Use Case Name	High Priority Complaint Alert
Primary Actor	n8n Automation
Description	System instantly alerts responsible teams about high-priority complaints.
Preconditions	Complaint priority = High.
Trigger	High-priority complaint created.
Main Flow	1. Complaint created. 2. n8n detects high priority. 3. Alert sent to assigned team.
Alternative Flow	Team unavailable → Escalation occurs.
Postconditions	Team notified immediately.
Business Rules	High-priority alerts must be instant.
Acceptance Criteria	Team receives alert immediately.